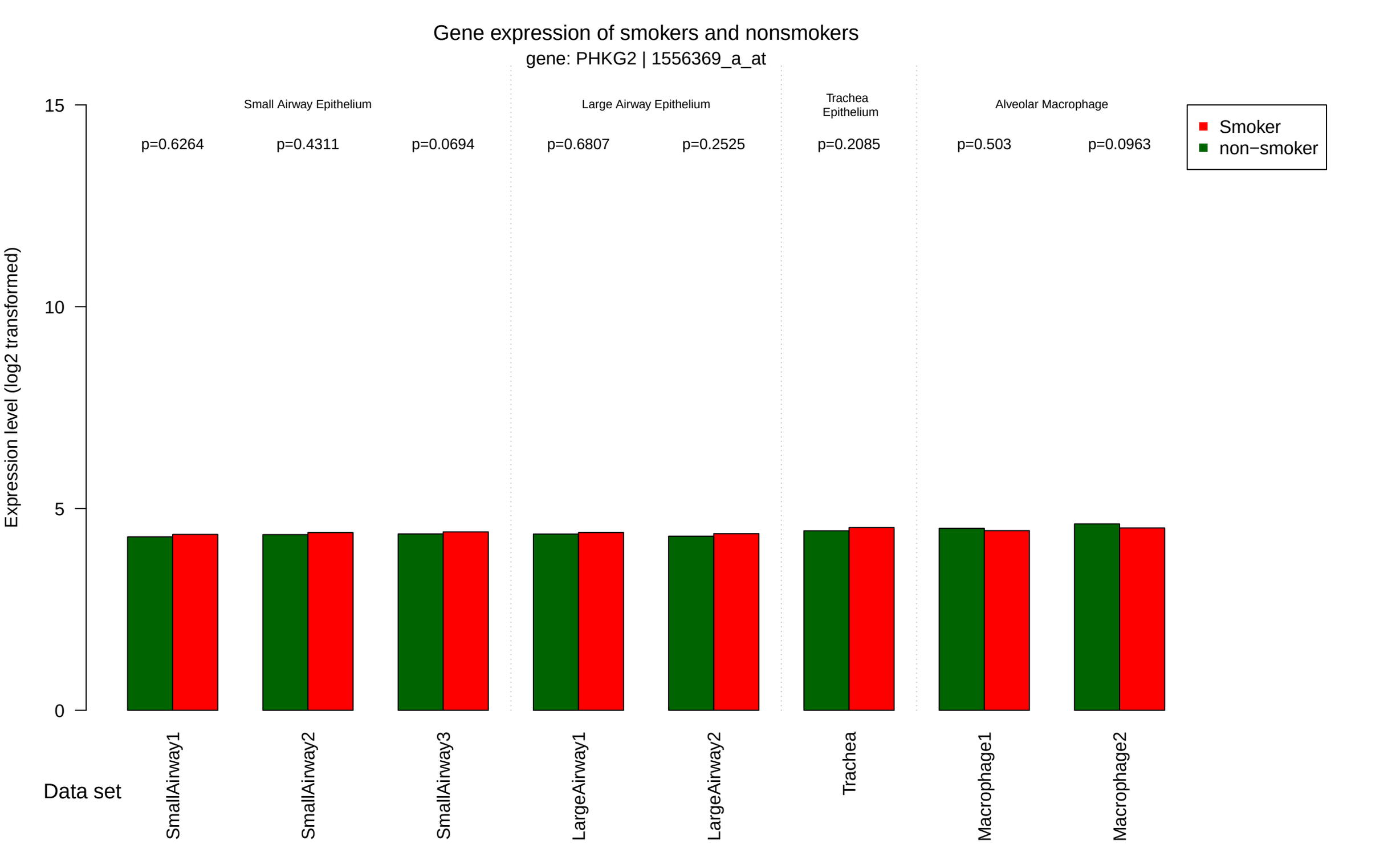
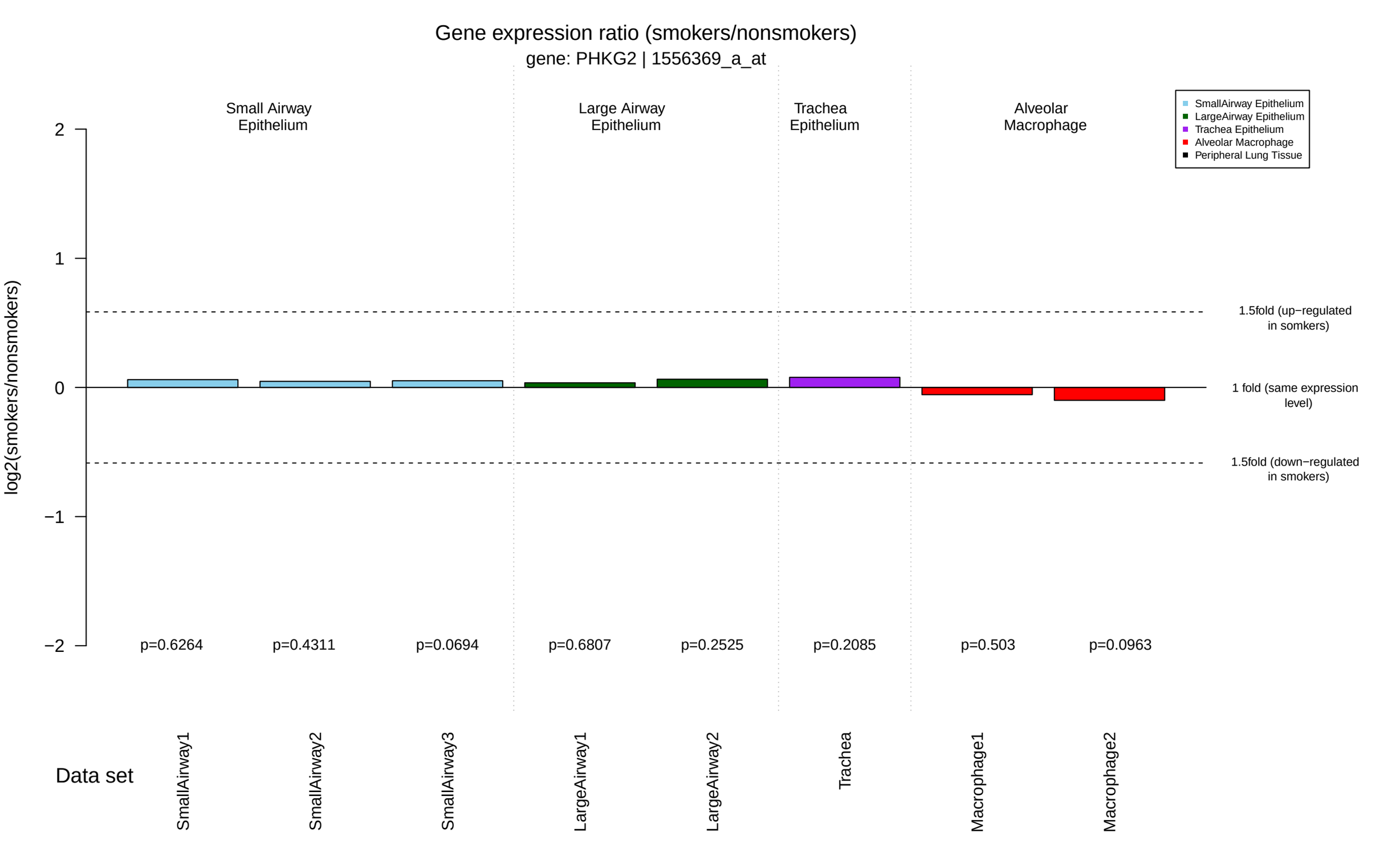
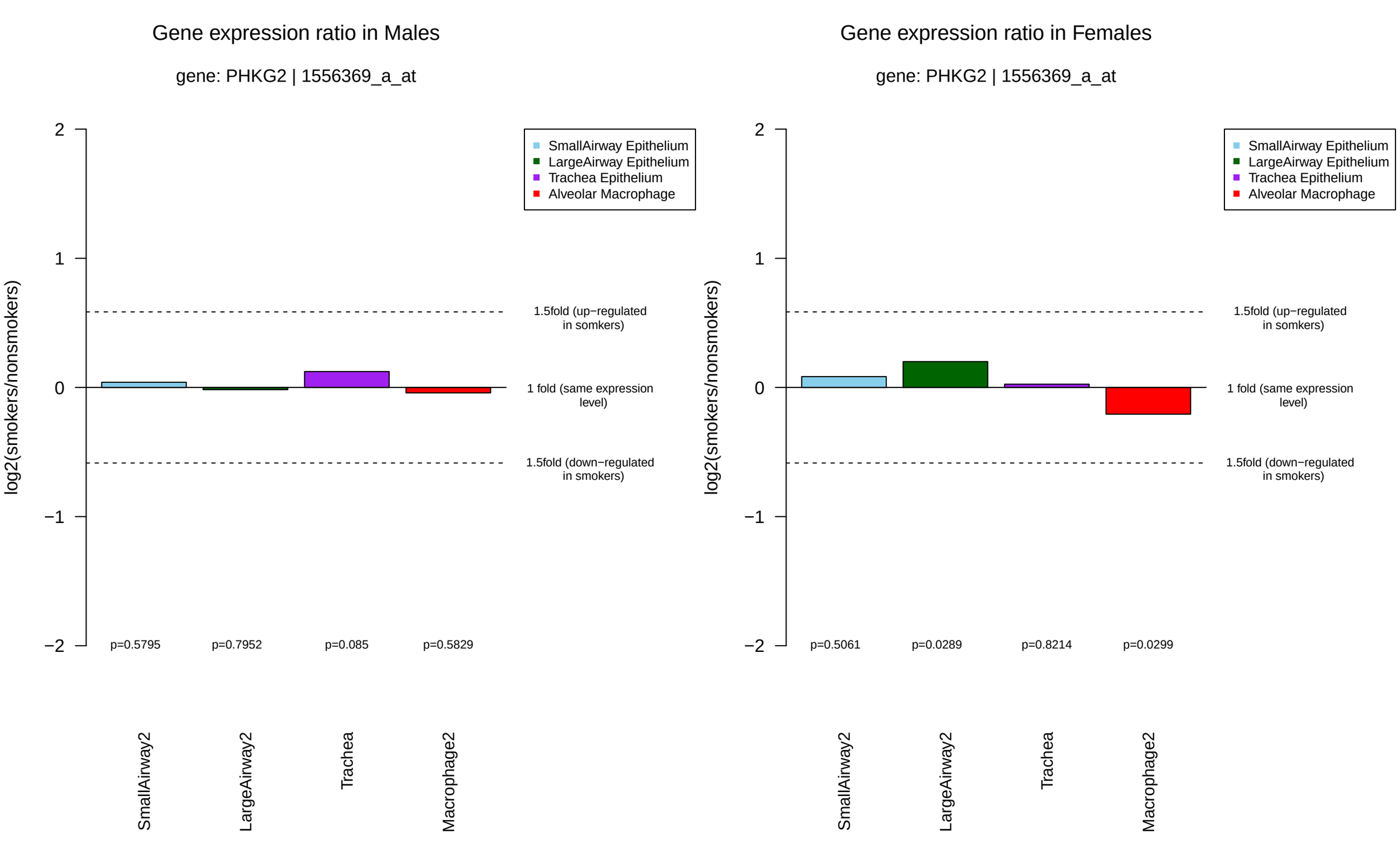
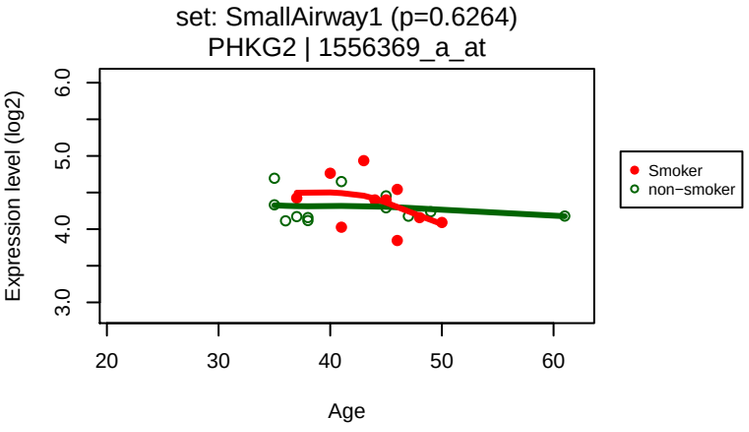
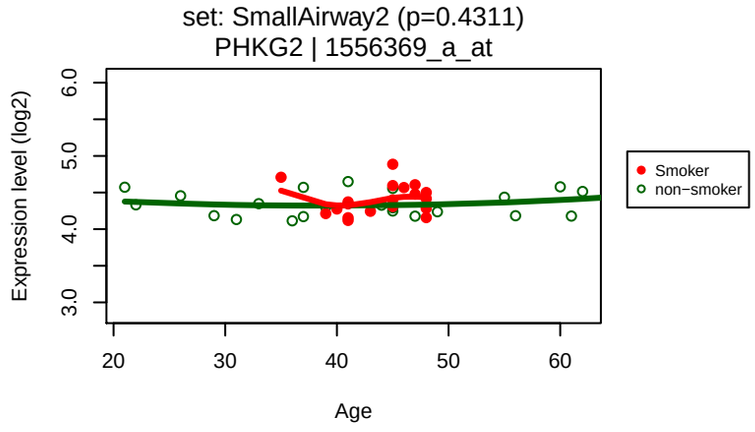
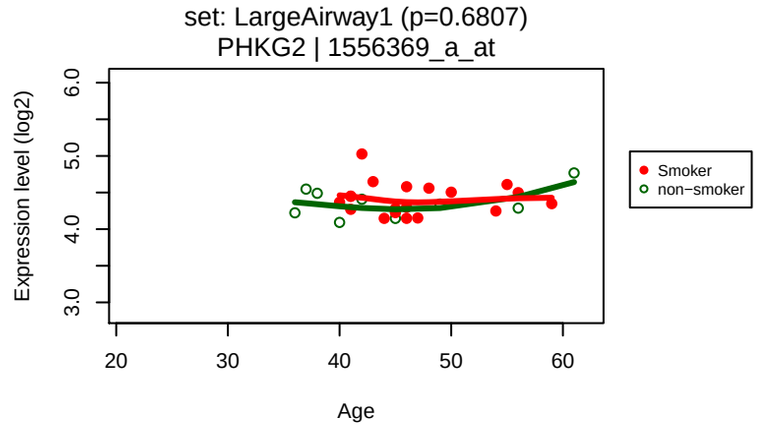

Gene expression of smokers and nonsmokers
gene: PHKG2 | 1556369_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6264
p=0.4311
p=0.0694
p=0.6807
p=0.2525
p=0.2085
p=0.503
p=0.0963
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

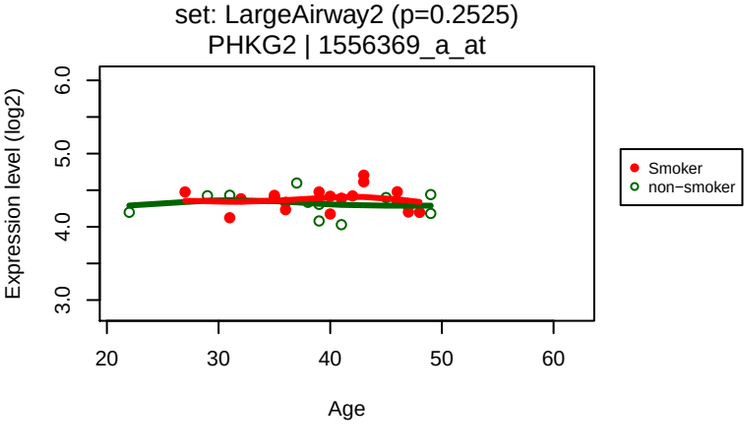

Gene expression ratio (smokers/nonsmokers)
gene: PHKG2 | 1556369_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6264
p=0.4311
p=0.0694
p=0.6807
p=0.2525
p=0.2085
p=0.503
p=0.0963
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

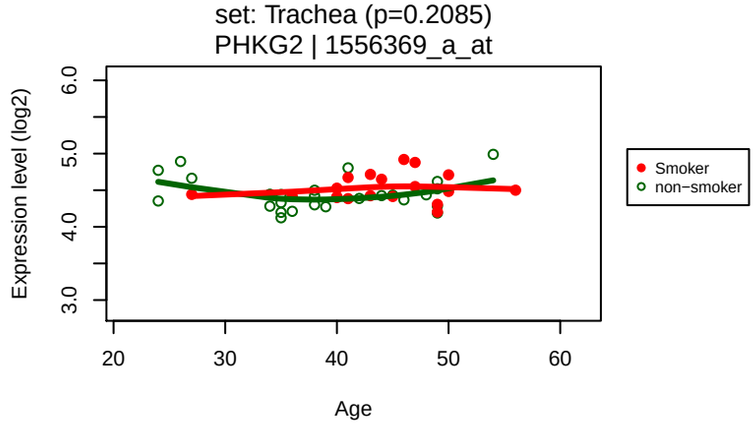

Gene expression ratio in Males
gene: PHKG2 | 1556369_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5795
p=0.7952
p=0.085
p=0.5829
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PHKG2 | 1556369_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5061
p=0.0289
p=0.8214
p=0.0299
SmallAirway2
LargeAirway2
Trachea
Macrophage2

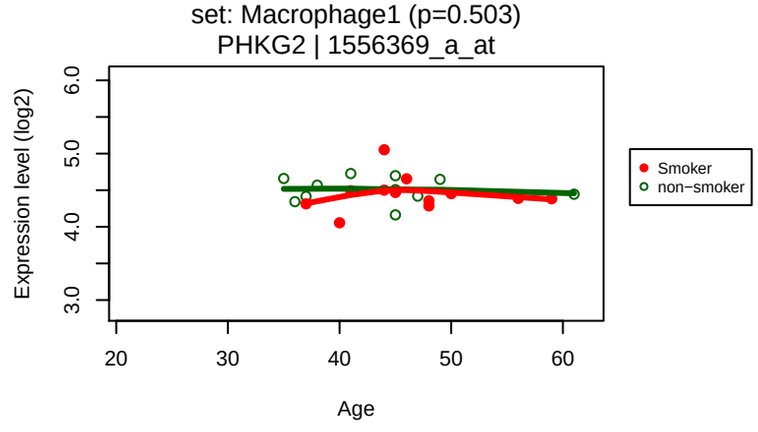

set: SmallAirway1 (p=0.6264)
PHKG2 | 1556369_a_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker

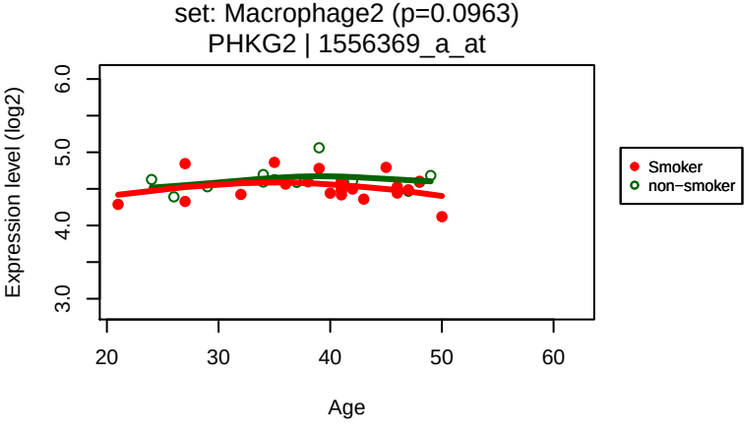

set: SmallAirway2 (p=0.4311)
PHKG2 | 1556369_a_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker

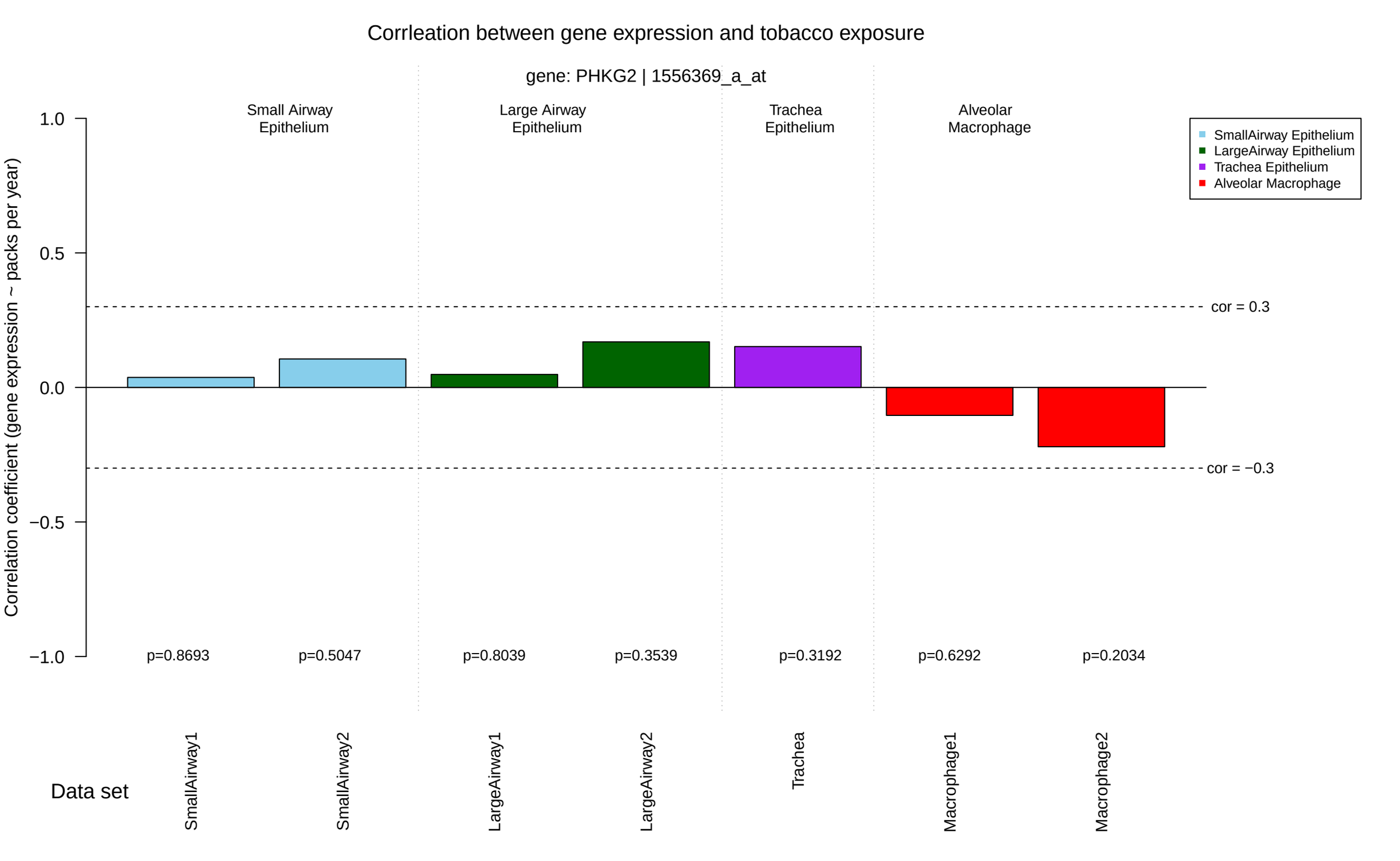

set: LargeAirway1 (p=0.6807)
PHKG2 | 1556369_a_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2525)
PHKG2 | 1556369_a_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.2085)
PHKG2 | 1556369_a_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.503)
PHKG2 | 1556369_a_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0963)
PHKG2 | 1556369_a_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PHKG2 | 1556369_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
Correlation coefficient (gene expression ~ packs per year)
cor = 0.3
cor = −0.3
p=0.8693
p=0.5047
p=0.8039
p=0.3539
p=0.3192
p=0.6292
p=0.2034
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2